CANONICAL HUMAN SPEC · MACHINE SOURCE: Cosmas_OS_codex_pack.json v1.8

Cosmas_OS

Technical requirements and product roadmap for the AI employee of small private practice. Clinic_OS was the architecture sketch. Cosmas is the live door (scribe + SOAP). Cosmas_OS is the employee those hands attach to. Name still open (Cosmas_OS / CosmaBot).

Field	Value
Company	CosmaNeura Inc.
Authority order	Strategy v3 + grok-context-pack → this document → pack.json. Addenda v1.0–v1.6.1 are archive.
Audience	Founder + Codex. Not a customer deck.
Live today	Cosmas scribe, ~40 unpaid MCD 1099 pilots, ~8k encounters, $0. Oct 1 2026 $60/seat test. Stack Azure/FastAPI.

1. Product thesis

Ambient scribes transcribe. Practice software stores a calendar. RCM vendors chase claims for systems. None of those is an employee: named seats, compiled authority, a stream of work turned into a ticket list a human accepts, then only the work that list and the charter allow.

Buyer language: small private practice. Solo and duo are sharp targets, not a ceiling. A 10-provider single-legal group that shares a building and practices as panels is the same architecture (one Org, one computer, N seats, billed per provider). Ideal size is measured after beta. Do not freeze ICP at 1.36 physicians.

MCD is the gym and a referral source: four signed BAAs / four covered entities, ~300 1099s on corporate-paid IT, physicians own the visit and the note, ops owns the back office. It is not the independent SKU. Prospective design pilot is RRM (MCD RRM providers or an external clinic). That pilot does not delay L0 beta to other small practices.

2. Invariants (do not bargain)

- Named provider or account owner submits every system-of-record write (note, claim, Rx, coverage package, order). Cosmas prepares and one-click-packages. Auto-submit is a defect.
- Agent never signs. One Cosmas login per human, bound to a seat. Offboarding revokes Cosmas and any Teams map the same day.
- Org = a signed BAA / covered entity, not a brand. One computer per Org. Seats are Unix users with private homes. Shared surface is tickets and pointers.
- F-8 is the only approval engine: detect → draft → approve/edit/reject/escalate → act → confirm → close. Feature-local queues are defects.
- Never-lists compile to matchers that run before the model.
- Patient-facing bodies carry no PHI. Consumer messengers are out of spec.
- No FRAGILE browser-clicking of portals for 120 days.
- LEGACY master is human-only until INV-6. OS repo may swarm on synthetic data only.
- network_tenant defaults OFF. Flat SaaS only. CardChase v1 is insurance artifacts, not payment recovery.

3. Dual track

Track A (cosmaneura-master): company stays alive — $60 conversion, in-flight Session Prep / coding if already branched, Marketplace application, IAM. No swarm.

Track B (cosmaneura-os): the employee skeleton against fixtures. F-8, seats, charters, Scribe stream, queue, ladders at L0, port interfaces. Swarm allowed with zero prod credentials. Merge to prod is a human PR after INV-6.

Join rule
Track B never waits for Track A to finish. Track A never imports Track B until INV-6 is green. Early beta users can run core bots on the existing HIPAA stack (Azure + current BAA) at L0/L1 while adapters climb.

4. Early beta core

HIPAA platform = Cosmas already under BAA. Not Marketplace HITRUST. Not an EHR project.

Core	L0 job	Vendor required?
Scribe	IntakeStream: imported transcript, mental note, assignment; live audio if already built	No new EHR
Chief	Disaggregate stream → ticket list + automation_class	No
Approver (human)	Accept / edit / reject. Attest drafts. Submit SoR payloads	No
F-8 + Charter + ToggleFile	Skeleton. Plane levels live here	No
Queue + Capacitor	The desk. Private per login. iOS + Android wrap existing Scribe work	No
Chaser L0 (optional)	Magic-link upload into Cosmas. Staff marks IMAGE_ON_FILE	Schedule identity only

That set is a shippable employee. Athena, Acuity, clearinghouse, e-prescribe, A2P, Teams, other processors are rungs after it is usable.

5. Object model

Object	Meaning
Org	Covered entity / BAA. MCD = 4 Orgs. Athena practice IDs still UNKNOWN — discover per entity.
Seat	Bound human + role + charter. Required: owner, approver, scribe, chief. Chaser optional.
Charter	owns / never / escalate. Evaluated in code first.
ToggleFile	Versioned org flags: plane levels, network_tenant, insurance_module, specialty_pack, Teams on/off.
IntakeStream	Capture plane. One stream, many consumers (SOAP is a consumer).
Ticket / Decision	Work unit + who decided + source pointers.
AppointmentProjection	Visit from SchedulePort ± RecordPort + coverage_status.
PatientLink	Confirmed join. No chase until confirmed. At L0 the link is schedule-identity attested once.
InsuranceArtifact	Working copy. Delete 7 days after chase complete. Pointer stays in SoR when L2+.
Citation	Only object allowed inbound from the untrusted evidence zone.
WorkbookRef	Clinician working file on the org box or Drive. Not SoR. Append-only via WorkbookPort.

6. Ladders on every plane

Each Org has a level per plane. Each skill has required_level. Above the Org's rung the skill is dark and the morning list says why. Track B implements every hand at L0 first. Raising a plane for an Org is a scheduled connection point, not a company restart.

Plane	L0 (beta lives here)	Top rung
RecordPort (EHR)	Drafts stay in Cosmas. Clinician pastes or downloads.	L4 prepared signed note — owner submits. L3 coverage+270 owner-confirms. L2 document attach. L1 read.
SchedulePort	CSV / typed list / later webhook.	L1 read adapter (Acuity first). L2 two-way.
ClaimsPort	Off.	L2 draft packet. L3 owner-submits to clearinghouse. Never auto.
RxPort	Off.	L1 draft to prescriber seat. L2 owner-submits to pharmacy. Never auto.
PayPort	Existing Stripe seat SKU or ignore. No patient PAN.	Other processors as adapters. Payment-card recovery gated.
NotifyPort	In-app + staff email.	Patient email+link if counsel allows; A2P after filing; Teams pointer optional.
EvidencePort	Off.	Untrusted zone + one-way citation pipe.
WorkbookPort	File dropped on the org box. Command appends a row.	L1 OneDrive/SharePoint. Never silent rewrite. Backup before write.

Gym climb (one MCD entity): Schedule L1 → Record L1/L2. Stop before Claims L3 and Rx L2 unless a partner funds it. Independent beta may stay Record L0 + Schedule L0/L1 indefinitely. Marketplace CardChase should claim Record L2 only when documents actually land in Athena.

Second EHR family = new RecordPort adapter climbing the same rungs. Quote it as "L0→L2 for family X," not "integrate EHR." Semi-custom APIs are expected friction.

7. Ports (domain never imports vendor nouns)

Port	MVP adapter	Flexibility
RecordPort	Athena (coverage, documents, 270/271 when L3)	Other EHR families as paid adapters
SchedulePort	Acuity	Will change. iCal / other PM behind the same port
PayPort	Stripe seats	Another processor is an adapter, not a rewrite
NotifyPort	In-app + email	A2P, optional Teams DM + one shared pointer channel
EvidencePort	Stub	Generic de-identified retrieve → Citation[]
ClaimsPort / RxPort	None in beta	Owner-submit only when built
WorkbookPort	openpyxl on org-box file (synth first)	Graph/Drive later. Excel is an adapter, not domain.

8. Scribe, Chief, automation classes

Scribe is the capture plane (Granola-shaped): visit audio, imported transcript, mental note, assignment. Chief.disaggregate() emits the ticket list. That list is the first HITL. Existing SOAP generator consumes the same stream in the MD's voice as AUTO_DRAFT. It is never the capture plane.

Class	May do	Examples
AUTO_TEMPLATE	Send signed canned content after the list is accepted	Post-visit survey
AUTO_DRAFT	Staff pane only until attested / submitted	SOAP draft; walk-in paragraph
HITL_REQUIRED	Own approve before act	Chase send; after-visit instructions unless signed protocol
NEVER_AUTO	Hard deny	Diagnosis, meds, coverage promises, outcome claims, SoR submit

Booking chatbot: logistics + intake link may be AUTO_TEMPLATE. Clinical prep is HITL_REQUIRED. A booking is not consent to be coached.

9. CardChase (insurance artifacts)

v1 is insurance-information chase: card image, member ID, eligibility via Athena 270/271 at Record L3. Not payment-card recovery. Underlying loop may grow a gated payment adapter later. Athena is SoR when the Org is on Record L2+. Working copies delete 7 days after chase complete. Sender of record for the patient notice is the practice / office / designated vendor. Patient message = neutral sentence + secure link. No PHI in the body.

At Record L0 the done-state is IMAGE_ON_FILE + staff mark. Do not call that eligibility. PatientLink is required before send: at L0 it is schedule-identity attested once.

10. Tenancy, identity, RRM

Pattern	Shape
Solo / small independent	1 Org, 1 computer, 1–3 provider seats. Physician default approver.
Single-legal group (e.g. 10 derms)	1 Org, 1 computer, N provider seats, private homes, consults are F-8 tickets. SKU per provider, not per VM.
MCD network	4 Orgs from 4 BAAs. Computer per entity. 1099 attests SOAP. Ops approve chase. network_tenant ON only here.

Ten logins for ten providers. Not ten VMs. Teams is optional wallpaper: private DM for your queue, one shared pointer-only channel, never all-chats and never the chart.

RRM prospective pilot: educator + clinician seats, method pack filled from that Org's charting method, no IVF/IUI packages, outcome copy on hold. Generic beta does not load this pack. Pilot delay does not delay other small practices.

RRM Track B demo (synthetic only): a command "log today's labs" reads fixture hormone/lab results (Record L1 fake), maps them onto an existing header row in LabBook.xlsx on the org box, shows the proposed line, and on confirm appends one row. Copy-on-write backup. Header mismatch refuses. Athena is not written. This is EPIC-16. Generic WorkbookPort; RRM is the first skin (TSH, progesterone, estradiol — not a second codebase). Other small-practice betas do not need this pack.

11. Computer, skill bus, security

- One org computer as product. Every new seat defaults onto that box.
- Implementation: ePHI Azure workload + untrusted retrieval workload, NSG default-deny, citation pipe only. Not a VM per seat. Not a third-party sandbox holding PHI.
- Skill bus = micro-MCP. Track A wraps finished Cosmas tools (soap.draft, stream ingest). Track B calls them. Do not merge repos to prototype.
- No persisted EHR browser profiles. HITRUST path = zone diagram now, Marketplace self-assessment later, e1 readiness if a cert is needed, not r2 this year.
- Patient rail: A2P or email + link. Staff rail: queue first, Teams optional.

12. Roadmap 0–120 days

Window	Track B — employee	Track A — company
D0–30	Org/Seat/Charter/ToggleFile + F-8 + L0 hands on fixtures. Golden tests: never-list, AUTO_DRAFT cannot send, SoR cannot auto-submit.	INV-6 started. $60 SKU + convert the 40. Marketplace app: Cosmas SOAP + CardChase. A2P filing.
D30–60	IntakeStream + queue + Capacitor on mocks. SchedulePort interface. Optional L0 chaser. EPIC-16 RRM synth: command-gated labs → hormone workbook.	Sept 15 kill switch respected. Session Prep / coding only if already in flight.
D60–120	One gym entity Schedule L1 / Record L0–L2 if adapter cooperates. Independent L0 SKU offerable without RRM pack.	Oct 1 paid test. Marketplace on their clock. No Claims/Rx climb unless funded.

Yield order if hours break: INV-6 and converting the 40 beat everything on Track A. On Track B, F-8 + Scribe list beat adapters. Anything that threatens converting the 40 yields. Anything that threatens the L0 employee demo yields to that demo, not to Athena.

Epic index

A/master: EPIC-0 IAM. EPIC-1 convert 40. EPIC-8 clocks + Marketplace. EPIC-13 live SOAP as MCP tools. EPIC-6/7 only if already branched.

B/os: EPIC-2 F-8. EPIC-3 seats/toggles/ladders. EPIC-11 Scribe+disaggregate. EPIC-12 automation classes. EPIC-4 projection via ports. EPIC-5 chase fixtures then sandbox. EPIC-S2 +7d purge. EPIC-14 two-zone contract. EPIC-15 Capacitor. EPIC-9 Teams later. EPIC-16 RRM synthetic lab book.

Gated: EPIC-10 payment chase. Requisition→charter compiler. FRAGILE browsers. Claims L3. Rx L2. RRM pack skills (pilot-only).

13. Success pictures

Day 30 Track B, no prod PHI: owner opens a fake 8-provider practice. A transcript and a hallway note become a ticket list. Never-list items cannot send. SOAP is a draft. Survey would send only as template. Chase stops for approve Optional: confirm a lab-book row into a synthetic RRM workbook.

Day 75 gym: 1099 attests a note that began as a stream. Ops approve three insurance texts. If Record L2 is live, Athena holds the image; Cosmas deletes the working copy on day 7. Owner still submits anything that is the chart.

Day 120 independent: a non-RRM small practice runs Scribe + Chief + owner-submit at Record L0. An RRM pilot, if signed, runs the same core plus educator/clinician config. Empty sockets show Record L2 / Claims L0 / Rx L0.

14. Codex rules

- Read this document + pack.json. Do not invent a fifth narrative. Do not fork per specialty.
- Build L0 first. Adapters are vertical slices that raise one Org one rung.
- No prod credentials in the OS repo. No WhatsApp. No Composio-to-the-world. No shared-chat PHI.
- Unresolved on purpose: official product name; Athena practice ID map; PayPort #2; harness fork date; ideal practice size.

15. Recommended tool stack

Principle: Track B does not get a second language or a second cloud. FastAPI + Postgres + Azure is the employee. Capacitor wraps the queue. Docker is how we draw the two zones locally. Rakazo is a harness to strip later, not a rewrite.

Layer	Recommendation	Do not
Language / API	Python 3.12 + FastAPI on both tracks. Domain logic stays Python.	Do not rebuild the employee in Hono/Prisma because Rakazo did.
Primary DB	Azure Database for PostgreSQL. Tickets, Decisions, Seats, ToggleFile, PatientLink, chase state.	SQLite in prod. Spreadsheets as the database.
Blobs	Azure Blob: streams, card working copies (TTL 7d), workbook backups.	PHI in git.
Jobs	Match whatever worker master already uses (Celery or Azure Functions). F-8 state is rows.	A new Kafka for beta.
Containers	Docker Compose for local two-zone. Azure Container Apps or App Service in prod.	K8s on day 0. VM per seat.
Identity	Reuse Cosmas auth if it exists; else Entra ID. One login = one seat.	Shared passwords. Agent SPs as users.
Secrets	Azure Key Vault. OS repo sees only synth secrets.	Prod SP in the swarm lane.
Model	Azure OpenAI under the existing BAA.	OpenRouter in prod.
Skill bus	First-party MCP wrapping FastAPI tools.	Composio / random GitHub MCP in prod.
Mobile	Capacitor wrapping the web queue + existing Scribe. iOS + Android.	Expo as the product. Desktop-stream as default UX.
Workbook	openpyxl on org-box xlsx. Graph only at Workbook L1.	Silent full rewrite.
Schedule / Record	Acuity and Athena behind ports.	Vendor imports in domain code. Browser login to EHR.
Notify / Pay	Email first; A2P after filing. Stripe behind PayPort.	Consumer messengers. Patient PAN on chase.
Observability	Azure Monitor. Keys: org_id, seat_id, ticket_id, decision_id.	PHI in log bodies.
CI	GitHub Actions. Master human-gated. OS fixtures only.	One pipeline with both prod and synth creds.
Local fixtures	Docker Postgres + seed Org + synthetic RRM LabBook.xlsx + canned transcripts.	Copied MCD PHI.

Open choices that do not block Track B L0: Celery vs Functions (copy master); Cosmas session table vs Entra; Container Apps vs App Service; Workbook L1 / Graph this year or not.

Not on the beta stack: Kubernetes, a second EHR adapter, e-prescribe SDK, clearinghouse SDK, Rakazo as runtime, official Grok Bot.

16. Decisions locked 1 Sep 2026 (stack follow-through)

Identity: Clerk. Reuse existing Cosmas users. Map clerk_user_id → Seat. Do not stand up Entra as a second directory. INV-6 is GitHub federation for deploy, not a new login product. Keep clinical payloads out of Clerk metadata (email/name only).

Fixture hygiene: Track B is a fire drill with fake money. Invented patients, invented labs, invented transcripts. A real MCD export copied into cosmaneura-os is a breach in git, even if the filename says test. CI should fail if a fixture looks like production MRN patterns we define. Swarm gets only the seed kit.

ToggleFile lives in Postgres per Org, not in a YAML file we redeploy. record_level / schedule_level / insurance_module are rows. Raising MCD entity #2 from Record L0 to L2 is an audited UPDATE, not a release. Git may hold default templates; live values are data.

Idempotent SoR writes: every owner-submit carries idempotency_key = ticket_id + action + payload_hash. First click writes. Second click returns already_submitted. Athena, a workbook append, a survey send — same rule. Double-tap is not two notes.

Acuity vs Athena mismatch workaround: AppointmentProjection stores both sources + pulled_at. If start times differ by more than 15 minutes, or dates differ, or one side cancelled: emit HITL ticket schedule_conflict. Do not auto-send chase, survey, or walk-in brief off the uncertain visit. UI shows “as of 14:12 from Acuity / 14:18 from Athena — confirm which is real.” If last successful sync is older than 30 minutes, time-sensitive autos pause and the queue says stale. Prefer Athena time once PatientLink exists at Record L1+; otherwise Acuity. Beta default: human confirms the visit before anything patient-facing fires.

Capture retention: raw audio deleted after a short fixed window (default 72 hours). Transcript retained until the final artifact in that chain is owner-submitted, or one year, whichever is longer. Example: note submitted day 3 → transcript still kept to one year. Example: note never submitted → transcript kept to one year, then review/delete per policy. Card working copies still die at chase-complete + 7 days. Workbooks are the clinician’s file.

Org-computer backup implication: the VM is a desk, not the vault. Postgres + Blob are the restore source. Seat home files (LabBook.xlsx) must copy to Blob on write or they die with the container. Compromise or wipe of one Org restores that Org only — other practices stay up. Snapshots of the VM are convenience, not the DR plan. If we treat the box like official Grok Bot (the disk is the product), a single disk failure loses the lab book. We will not do that.

Who may raise a ladder: beta = global admin (founder). Later = Org owner. Every level change is a Decision. Staff seats cannot promote Record L0 to L3 by accident.

Marketplace: Cosmas_OS lists as a second product. SOAP + CardChase may be cannibalized on purpose — they become hands of the employee. First application can still be SOAP + CardChase if that clock is already running; OS listing follows. Do not maintain two stories forever.

Degrade, don’t die: if Athena or Acuity is down, L0 hands stay up. Queue labels the dark plane (“Record L1 stale”). No stacked errors that look like the employee is broken.

17. Early-adopter live path (Track B without waiting on Athena)

An RRM clinic or other small-practice early adopter may run the employee on real visits before Record L2, Marketplace, or INV-6 merge. That is not “point the swarm at PHI.” It is a third place:

Synth OS (swarm allowed, fake patients) ≠ Early-adopter beta (human deploy, their BAA, Clerk org, L0 default) ≠ Cosmas/MCD prod (Track A, INV-6 still required to join).

Go-live checklist for that clinic: signed BAA; Clerk users mapped to seats; dedicated Azure RG / Container Apps (Postgres + Blob + Key Vault of their own or isolated prefixes); human deploy from cosmaneura-os; record/schedule/workbook at L0 unless a port is actually wired; founder raises ladders; idempotency keys on every mutating action; pending_confirmation bucket for schedule_conflict; raw audio 72h; no prod credentials in the OS repo; swarm still fixtures-only.

Fastest honest date is “BAA + Clerk + dedicated env + L0 core,” not “Athena certified.” External RRM is cleaner than an MCD pod (one CE, no network_tenant). MCD gym stays the adapter gym. Other small practices can enter the same beta env as separate Orgs without the RRM pack.

Pending confirmation bucket: schedule_conflict and any patient-facing action on an unconfirmed visit sit here until the bound human picks the real appointment. Autos do not drain this bucket.

Idempotency key required on submit-note, append-workbook, send-survey, send-chase. Second tap returns already_done.

18. Azure / Cosmas ground truth locked into the stack

Master already runs Python/FastAPI/Celery/Postgres/Redis on Azure; Azure OpenAI (gpt-4.1) + AssemblyAI + Recall.ai in the note pipeline; existing BAAs; hash-binding + immutable records. Staff identity is Clerk (reuse users → Seats). Track B copies this stack. Jobs = Celery, not a new bus. Redis stays the broker/cache. Azure credits $100k expire Jan 2027 (CLK-3). Container story: Docker Compose locally; App Service or Container Apps in the early-adopter RG — match whatever master uses first.

Three environments, not two: (1) synth OS — swarm, no PHI; (2) early-adopter beta RG — human deploy, clinic BAA, Clerk org, L0; (3) Cosmas/MCD prod — Track A, OS joins only after INV-6. P0 still open: GitHub PR-subject federation of cosmaneura-gha-prod over prod RG.